

Credit Risk Measurement

Advanced Topics in Banking - Project 2

Group 4 | WU Wien, SS 2026

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```
knitr::opts_chunk$set(  
  echo = TRUE,  
  warning = FALSE,  
  message = FALSE,  
  fig.width = 7,  
  fig.height = 5,  
  fig.pos = "H",  
  out.width = "100%"  
)  
  
if(!require("pacman")) install.packages("pacman")
```

Loading required package: pacman

```
pacman::p_load(  
  data.table,  
  tidyverse,  
  knitr,  
  kableExtra,  
  ggplot2,  
  openxlsx,  
  dplyr  
)  
  
BASE <- "data/"
```

```
ratings <- read.xlsx(paste0(BASE, "Data_Group_4.xlsx"), sheet = "Ratings")
portfolio <- read.xlsx(paste0(BASE, "Data_Group_4.xlsx"), sheet = "Portfolio")
```

```
lapply(list(ratings, portfolio), setDT)
```

```
## [[1]]
##      Obligor 2024.12.31 2025.01.31 2025.02.28 2025.03.31 2025.04.30
##      <num>      <char>      <char>      <char>      <char>      <char>
## 1:         1      CCC/C      CCC/C      CCC/C          B          B
## 2:         2      BBB      BBB      BBB      BBB      BBB
## 3:         3      BB       BB       BB       BB       BB
## 4:         4      B        B        B        B        B
## 5:         5      AAA      AAA      AAA      AAA      AAA
## ---
## 9996:      9996      BB       BB       A        A        A
## 9997:      9997      BB       BB       BB      BB      BB
## 9998:      9998      CCC/C      CCC/C      CCC/C      CCC/C      CCC/C
## 9999:      9999      BB       BB       BB      BB      BB
## 10000:     10000      AA       AA       AA      AA      AA
##      2025.05.31 2025.06.30 2025.07.31 2025.08.31 2025.09.30 2025.10.31
##      <char>      <char>      <char>      <char>      <char>      <char>
## 1:         B      B        B        B        B        B
## 2:      BBB      BBB      BBB      BBB      BBB      BBB
## 3:      BB       BB       BB       BB       BB       BB
## 4:         B      B        B        B        B        B
## 5:      AAA      AAA      AAA      AAA      AAA      AAA
## ---
## 9996:         A      A        A        A        A        A
## 9997:      BB      BB       BB       BB       BB       BB
## 9998:      CCC/C      CCC/C      CCC/C      CCC/C      CCC/C      CCC/C
## 9999:      BB      BB       BB       BB       BB       BB
## 10000:      AA      AA       AA       AA       AA       AA
##      2025.11.30 2025.12.31
##      <char>      <char>
## 1:         B      B
## 2:      BBB      BBB
## 3:      BB       BB
## 4:         B      B
## 5:      AAA      AAA
## ---
## 9996:         A      A
## 9997:      BB      BB
## 9998:      CCC/C      CCC/C
## 9999:      BB      BB
## 10000:      AA      AA
##
```

[[2]]

##	Obligor	Rating	Industry	Loan.Volume	
##	<num>	<char>	<num>	<num>	
##	1:	1	BBB	1	169000
##	2:	2	BB	3	149000
##	3:	3	BB	2	850000
##	4:	4	BB	2	76000
##	5:	5	CCC/C	1	174000
##	6:	6	BBB	1	108000
##	7:	7	A	1	122000
##	8:	8	A	3	33000
##	9:	9	BB	3	22000
##	10:	10	BBB	3	38000
##	11:	11	AAA	1	142000
##	12:	12	BB	2	113000
##	13:	13	A	3	850000
##	14:	14	B	3	33000
##	15:	15	BB	2	166000
##	16:	16	BB	1	84000
##	17:	17	BBB	1	143000
##	18:	18	A	1	191000
##	19:	19	B	2	96000
##	20:	20	BB	3	90000
##	21:	21	BB	3	850000
##	22:	22	BBB	1	850000
##	23:	23	A	1	850000
##	24:	24	A	1	198000
##	25:	25	BB	3	162000
##	26:	26	BB	2	21000
##	27:	27	CCC/C	1	850000
##	28:	28	BBB	2	108000
##	29:	29	BBB	2	49000
##	30:	30	BB	3	50000
##	31:	31	BBB	1	48000
##	32:	32	A	1	33000
##	33:	33	BB	1	137000
##	34:	34	AAA	1	72000
##	35:	35	AA	3	103000
##	36:	36	AA	2	91000
##	37:	37	BBB	1	182000
##	38:	38	A	2	88000
##	39:	39	BBB	2	44000
##	40:	40	B	3	37000
##	41:	41	BB	3	52000
##	42:	42	BB	3	79000
##	43:	43	A	2	46000
##	44:	44	BB	3	144000
##	45:	45	A	3	94000

##	46:	46	A	1	143000
##	47:	47	AA	1	850000
##	48:	48	CCC/C	1	21000
##	49:	49	BBB	2	168000
##	50:	50	BBB	1	148000
##	51:	51	BBB	2	193000
##	52:	52	BBB	3	193000
##	53:	53	A	2	119000
##	54:	54	BBB	1	85000
##	55:	55	BBB	1	139000
##	56:	56	BBB	3	178000
##	57:	57	AA	2	21000
##	58:	58	BB	2	172000
##	59:	59	AA	1	31000
##	60:	60	CCC/C	1	52000
##	61:	61	A	2	125000
##	62:	62	BBB	1	34000
##	63:	63	AA	1	59000
##	64:	64	BB	2	44000
##	65:	65	AAA	1	65000
##	66:	66	BBB	2	91000
##	67:	67	BBB	1	850000
##	68:	68	BBB	1	56000
##	69:	69	B	2	49000
##	70:	70	A	1	57000
##	71:	71	BBB	3	97000
##	72:	72	BB	2	64000
##	73:	73	BBB	1	74000
##	74:	74	BB	3	149000
##	75:	75	BB	2	24000
##	76:	76	B	2	121000
##	77:	77	A	1	198000
##	78:	78	A	3	190000
##	79:	79	A	3	93000
##	80:	80	BBB	3	110000
##	81:	81	B	2	137000
##	82:	82	BB	2	151000
##	83:	83	CCC/C	1	111000
##	84:	84	BBB	3	94000
##	85:	85	BB	3	200000
##	86:	86	CCC/C	1	113000
##	87:	87	BB	1	135000
##	88:	88	AA	1	60000
##	89:	89	BBB	2	103000
##	90:	90	B	1	67000
##	91:	91	BBB	3	166000
##	92:	92	AA	2	195000
##	93:	93	B	1	184000

```
## 94:      94      BB      1      186000
## 95:      95     BBB      1      139000
## 96:      96       B      3      163000
## 97:      97     BBB      1       21000
## 98:      98      BB      3      138000
## 99:      99      BB      3      120000
## 100:     100     BBB      2      143000
##      Obligor Rating Industry Loan.Volume
##      <num> <char>      <num>      <num>
```

```
lapply(list(ratings, portfolio), str)
```

```
## Classes 'data.table' and 'data.frame':  10000 obs. of  14 variables:
## $ Obligor   : num  1 2 3 4 5 6 7 8 9 10 ...
## $ 2024.12.31: chr  "CCC/C" "BBB" "BB" "B" ...
## $ 2025.01.31: chr  "CCC/C" "BBB" "BB" "B" ...
## $ 2025.02.28: chr  "CCC/C" "BBB" "BB" "B" ...
## $ 2025.03.31: chr  "B" "BBB" "BB" "B" ...
## $ 2025.04.30: chr  "B" "BBB" "BB" "B" ...
## $ 2025.05.31: chr  "B" "BBB" "BB" "B" ...
## $ 2025.06.30: chr  "B" "BBB" "BB" "B" ...
## $ 2025.07.31: chr  "B" "BBB" "BB" "B" ...
## $ 2025.08.31: chr  "B" "BBB" "BB" "B" ...
## $ 2025.09.30: chr  "B" "BBB" "BB" "B" ...
## $ 2025.10.31: chr  "B" "BBB" "BB" "B" ...
## $ 2025.11.30: chr  "B" "BBB" "BB" "B" ...
## $ 2025.12.31: chr  "B" "BBB" "BB" "B" ...
## Classes 'data.table' and 'data.frame':  100 obs. of  4 variables:
## $ Obligor   : num  1 2 3 4 5 6 7 8 9 10 ...
## $ Rating     : chr  "BBB" "BB" "BB" "BB" ...
## $ Industry   : num  1 3 2 2 1 1 1 3 3 3 ...
## $ Loan.Volume: num  169000 149000 850000 76000 174000 108000 122000 33000 22000 38000 ...

## [[1]]
## NULL
##
## [[2]]
## NULL
```

```
lapply(list(ratings, portfolio), head)
```

```
## [[1]]
##      Obligor 2024.12.31 2025.01.31 2025.02.28 2025.03.31 2025.04.30 2025.05.31
##      <num>      <char>      <char>      <char>      <char>      <char>      <char>
## 1:      1      CCC/C      CCC/C      CCC/C      B      B      B
## 2:      2      BBB      BBB      BBB      BBB      BBB      BBB
## 3:      3      BB      BB      BB      BB      BB      BB
## 4:      4      B      B      B      B      B      B
## 5:      5      AAA      AAA      AAA      AAA      AAA      AAA
## 6:      6      BBB      BBB      BBB      BBB      BBB      BBB
```

```
##      2025.06.30 2025.07.31 2025.08.31 2025.09.30 2025.10.31 2025.11.30 2025.12.31
##      <char>      <char>      <char>      <char>      <char>      <char>      <char>
## 1:      B        B        B        B        B        B        B
## 2:      BBB      BBB      BBB      BBB      BBB      BBB      BBB
## 3:      BB       BB       BB       BB       BB       BB       BB
## 4:      B        B        B        B        B        B        B
## 5:      AAA      AAA      AAA      AAA      AAA      AAA      AAA
## 6:      BBB      BBB      BBB      BBB      BBB      BBB      BBB
```

```
##
```

```
## [[2]]
```

```
##      Obligor Rating Industry Loan.Volume
##      <num> <char>      <num>      <num>
## 1:      1      BBB      1        169000
## 2:      2      BB       3        149000
## 3:      3      BB       2        850000
## 4:      4      BB       2        76000
## 5:      5      CCC/C    1        174000
## 6:      6      BBB      1        108000
```

```
lapply(list(ratings, portfolio), summary)
```

```
## [[1]]
```

```
##      Obligor      2024.12.31      2025.01.31      2025.02.28
## Min.      : 1      Length :10000      Length :10000      Length :10000
## 1st Qu.: 2501      N.unique : 7      N.unique : 8      N.unique : 8
## Median : 5000      N.blank : 0      N.blank : 0      N.blank : 0
## Mean    : 5000      Min.nchar: 1      Min.nchar: 1      Min.nchar: 1
## 3rd Qu.: 7500      Max.nchar: 5      Max.nchar: 5      Max.nchar: 5
## Max.      :10000
##      2025.03.31      2025.04.30      2025.05.31      2025.06.30
## Length :10000      Length :10000      Length :10000      Length :10000
## N.unique : 8      N.unique : 8      N.unique : 8      N.unique : 8
## N.blank : 0      N.blank : 0      N.blank : 0      N.blank : 0
## Min.nchar: 1      Min.nchar: 1      Min.nchar: 1      Min.nchar: 1
## Max.nchar: 5      Max.nchar: 5      Max.nchar: 5      Max.nchar: 5
##
##      2025.07.31      2025.08.31      2025.09.30      2025.10.31
## Length :10000      Length :10000      Length :10000      Length :10000
## N.unique : 8      N.unique : 8      N.unique : 8      N.unique : 8
## N.blank : 0      N.blank : 0      N.blank : 0      N.blank : 0
## Min.nchar: 1      Min.nchar: 1      Min.nchar: 1      Min.nchar: 1
## Max.nchar: 5      Max.nchar: 5      Max.nchar: 5      Max.nchar: 5
##
##      2025.11.30      2025.12.31
## Length :10000      Length :10000
## N.unique : 8      N.unique : 8
## N.blank : 0      N.blank : 0
## Min.nchar: 1      Min.nchar: 1
```

```
## Max.nchar:    5    Max.nchar:    5
##
##
## [[2]]
##      Obligor          Rating      Industry      Loan.Volume
## Min.   : 1.00    Length   :100    Min.    :1.00    Min.    : 21000
## 1st Qu.: 25.75    N.unique  : 7    1st Qu.:1.00    1st Qu.: 59750
## Median : 50.50    N.blank  : 0    Median :2.00    Median :112000
## Mean   : 50.50    Min.nchar: 1    Mean   :1.87    Mean   :165760
## 3rd Qu.: 75.25    Max.nchar: 5    3rd Qu.:3.00    3rd Qu.:166000
## Max.    :100.00                Max.    :3.00    Max.    :850000

nrow(ratings)

## [1] 10000

nrow(portfolio)

## [1] 100

print(colnames(ratings))

## [1] "Obligor"      "2024.12.31" "2025.01.31" "2025.02.28" "2025.03.31"
## [6] "2025.04.30" "2025.05.31" "2025.06.30" "2025.07.31" "2025.08.31"
## [11] "2025.09.30" "2025.10.31" "2025.11.30" "2025.12.31"

print(colnames(portfolio))

## [1] "Obligor"      "Rating"      "Industry"      "Loan.Volume"
# Correctly parse the dots and format to dashes
colnames(ratings)[-1] <- format(as.Date(colnames(ratings)[-1], format = "%Y.%m.%d"), "%Y-%m-%d")
print(colnames(ratings))

## [1] "Obligor"      "2024-12-31" "2025-01-31" "2025-02-28" "2025-03-31"
## [6] "2025-04-30" "2025-05-31" "2025-06-30" "2025-07-31" "2025-08-31"
## [11] "2025-09-30" "2025-10-31" "2025-11-30" "2025-12-31"
```

1 Task

1.1

Based on the rating history of 10,000 representative borrowers, determine the one-year migration matrix based on the Lando and Skodeberg procedure. In doing so, determine the migration matrix using the generator matrix based on at least the first five terms.

Estimation of the generator matrix come from this formula:

$$\lambda_{ij} = \frac{N_{ij}}{\int Y_i(s) ds} \quad i \neq j \quad \text{and} \quad \lambda_{ii} = - \sum_{j \neq i} \lambda_{ij}$$

where N_{ij} is the number of observed migrations from rating i to rating j and $Y_i(s)$ is the number of borrowers in rating i at time s . And the integral: $\int Y_i(s) ds$ is the total time spent by all borrowers

in rating i during the observation period. The one-year migration matrix can then be obtained by exponentiating the generator matrix: $W(\Delta t) = e^{\Lambda \cdot \Delta t} = \sum_{k=0}^{\infty} \frac{(\Lambda \cdot \Delta t)^k}{k!}$

```
ratings_order <- c("AAA", "AA", "A", "BBB", "BB", "B", "CCC/C", "D")
K <- length(ratings_order)
```

```
# melt ratings to long format
```

```
ratings_long <- melt(
  ratings,
  id.vars = "Obligor",
  variable.name = "date",
  value.name = "rating",
  variable.factor = FALSE
)
```

```
ratings_long[, date := as.Date(date)]
ratings_long[, rating := factor(rating, levels = ratings_order)]
```

```
setorder(ratings_long, Obligor, date)
```

```
nrow(ratings_long)
```

```
## [1] 130000
```

```
class(ratings_long$date)
```

```
## [1] "Date"
```

```
levels(ratings_long$rating)
```

```
## [1] "AAA" "AA" "A" "BBB" "BB" "B" "CCC/C" "D"
```

```
sum(is.na(ratings_long$rating))
```

```
## [1] 0
```

```
setdiff((unique(as.character(ratings_long$rating))), ratings_order)
```

```
## character(0)
```

```
head(ratings_long, 10)
```

Obligor	date	rating
1	2024-12-31	CCC/C
1	2025-01-31	CCC/C
1	2025-02-28	CCC/C
1	2025-03-31	B
1	2025-04-30	B
1	2025-05-31	B
1	2025-06-30	B

Obligor	date	rating
1	2025-07-31	B
1	2025-08-31	B
1	2025-09-30	B

```
ncol(ratings) - 1
```

```
## [1] 13
```

```
head(colnames(ratings), 25)
```

```
## [1] "Obligor"      "2024-12-31" "2025-01-31" "2025-02-28" "2025-03-31"
## [6] "2025-04-30" "2025-05-31" "2025-06-30" "2025-07-31" "2025-08-31"
## [11] "2025-09-30" "2025-10-31" "2025-11-30" "2025-12-31"
```

```
range(ratings_long$date)
```

```
## [1] "2024-12-31" "2025-12-31"
```

```
length(unique(ratings_long$date))
```

```
## [1] 13
```

```
# prev rating column
```

```
ratings_long[, prev_rating := shift(rating, n = 1L, type = "lag"), by = Obligor]
```

```
# drop first rows of each obligot
```

```
transitions <- ratings_long[!is.na(prev_rating)]
```

```
# cross tabulate: rows = from, cols = to, it means how many times we observed a transition from
```

```
N <- table(from = transitions$prev_rating, to = transitions$rating)
```

```
# convert from table to matrix
```

```
N <- unclass(N)
```

```
storage.mode(N) <- "double"
```

```
dim(N)
```

```
## [1] 8 8
```

```
rownames(N)
```

```
## [1] "AAA" "AA" "A" "BBB" "BB" "B" "CCC/C" "D"
```

```
colnames(N)
```

```
## [1] "AAA" "AA" "A" "BBB" "BB" "B" "CCC/C" "D"
```

```
sum(N)
```

```
## [1] 120000
```

```
diag(N) / rowSums(N)
```

```
##      AAA      AA      A      BBB      BB      B      CCC/C      D
## 0.9927156 0.9910590 0.9931906 0.9906120 0.9868227 0.9849699 0.9597836 1.0000000
```

```
N["D",]
```

```
##      AAA      AA      A      BBB      BB      B      CCC/C      D
##      0      0      0      0      0      0      0      0 1497
```

```
N
```

```
##      to
## from      AAA      AA      A      BBB      BB      B      CCC/C      D
## AAA      5860      37      4      1      1      0      0      0
## AA        7 11417      85      11      0      0      0      0
## A          1      39 23191     104      6      4      1      4
## BBB        1      11     120 35982     164     28      7     10
## BB         2      0      5     102 23440     163     22     19
## B          0      0      1      2     52 11927     62     65
## CCC/C      0      0      4      6      7     45 5322    161
## D          0      0      0      0      0      0      0 1497
```

```
# exact days between ratings
```

```
ratings_long[, prev_date := shift(date, n = 1L, type = "lag"), by = Obligor]
ratings_long[, dt := as.numeric((date - prev_date) / 365.25)]
```

```
# tapply to apply sum to dt by prev_rating, so we get the total time spent in each rating category
```

```
# tapply arguments: 1) the vector to apply the function to, 2) the factor to group by, 3) the function
```

```
Y <- tapply(ratings_long$dt, ratings_long$prev_rating, sum, na.rm = TRUE)
```

```
Y <- Y[ratings_order] # reorder Y according to ratings_order
```

```
Y # total time spent in each rating category
```

```
##      AAA      AA      A      BBB      BB      B      CCC/C      D
## 491.4935 959.3292 1944.6434 3024.5667 1977.8426 1008.5394 461.5140 125.2266
```

```
length(Y) # number of rating categories
```

```
## [1] 8
```

```
sum(Y) # total time spent in all rating categories
```

```
## [1] 9993.155
```

```
Y / sum(Y) # proportion of time spent in each rating category
```

```
##      AAA      AA      A      BBB      BB      B      CCC/C
## 0.04918301 0.09599863 0.19459753 0.30266384 0.19791973 0.10092301 0.04618301
##      D
```

```
## 0.01253123
```

```
# check: total time spent in all ratings should be equal to the total time between the first and last rating
cbind(N_rowsum = rowSums(N), Y_times_12 = Y * 12)
```

```
##      N_rowsum Y_times_12
## AAA      5903   5897.922
## AA      11520  11511.951
## A       23350  23335.721
## BBB     36323  36294.801
## BB      23753  23734.111
## B       12109  12102.472
## CCC/C    5545   5538.168
## D        1497   1502.719
```

```
# off diagonal elements of the generator matrix, divide the number of transitions by the total time
Lambda <- sweep(N, MARGIN = 1, STATS = Y, FUN = "/")
```

```
diag(Lambda) <- 0
```

```
diag(Lambda) <- -rowSums(Lambda)
```

```
rowSums(Lambda)
```

```
##      AAA      AA      A      BBB      BB
## 8.673617e-18 -1.734723e-18 9.540979e-18 1.084202e-17 -3.469447e-18
##      B      CCC/C      D
## 0.000000e+00 0.000000e+00 0.000000e+00
```

```
all(abs(rowSums(Lambda)) < 1e-10)
```

```
## [1] TRUE
```

```
all(Lambda[row(Lambda) != col(Lambda)] >= 0)
```

```
## [1] TRUE
```

```
all(diag(Lambda) <= 0)
```

```
## [1] TRUE
```

```
Lambda["D",]
```

```
##      AAA      AA      A      BBB      BB      B CCC/C      D
##      0      0      0      0      0      0      0      0
```

```
expm_taylor <- function(M, k_max) {
```

```
  K <- nrow(M)
```

```
  P <- diag(K)
```

```
  term <- diag(K)
```

```
  for (k in 1:k_max) {
```

```
    term <- (term %*% M) / k
```

```

      P <- P + term
    }
    dimnames(P) <- dimnames(M)
  P
}

P5 <- expm_taylor(Lambda, k_max = 5)
P20 <- expm_taylor(Lambda, k_max = 20)

```

```
rowSums(P5)
```

```
##   AAA   AA   A   BBB   BB   B CCC/C   D
##   1     1   1     1     1   1     1   1
```

```
rowSums(P20)
```

```
##   AAA   AA   A   BBB   BB   B CCC/C   D
##   1     1   1     1     1   1     1   1
```

```
max(abs(rowSums(P20) - 1))
```

```
## [1] 4.440892e-16
```

```
range(P5)
```

```
## [1] 0 1
```

```
range(P20)
```

```
## [1] 0 1
```

```
P20["D",]
```

```
##   AAA   AA   A   BBB   BB   B CCC/C   D
##   0     0   0     0     0   0     0   1
```

```
max(abs(P5 - P20))
```

```
## [1] 1.955294e-05
```

```
PD_raw <- P20[, "D"]
```

```

PD <- pmax(PD_raw, 0.0005) # set a floor for PD to avoid zero probabilities
data.frame(rating = names(PD), PD_raw, PD)

```

	rating	PD_raw	PD
AAA	AAA	0.0000271	0.0005000
AA	AA	0.0001145	0.0005000
A	A	0.0022365	0.0022365
BBB	BBB	0.0041178	0.0041178
BB	BB	0.0131906	0.0131906
B	B	0.0679631	0.0679631

	rating	PD_raw	PD
CCC/C	CCC/C	0.2795639	0.2795639
D	D	1.0000000	1.0000000

```
round(P20, 6)
```

```
##           to
## from      AAA      AA      A      BBB      BB      B      CCC/C      D
##  AAA  0.916483 0.068399 0.010571 0.002532 0.001877 0.000095 0.000015 0.000027
##  AA   0.006645 0.899273 0.080905 0.012437 0.000444 0.000147 0.000035 0.000114
##  A    0.000550 0.018366 0.923286 0.048762 0.004089 0.002196 0.000515 0.002237
##  BBB  0.000346 0.003634 0.036263 0.895622 0.047708 0.010075 0.002234 0.004118
##  BB   0.000904 0.000146 0.003246 0.045310 0.856744 0.070349 0.010110 0.013191
##  B    0.000024 0.000017 0.001193 0.003208 0.044031 0.838978 0.044586 0.067963
##  CCC/C 0.000011 0.000091 0.006858 0.010327 0.013291 0.070906 0.618952 0.279564
##  D     0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 0.000000 1.000000
```

```
pacman::p_load(expm)
P_expm <- expm::expm(Lambda)
max(abs(P_expm - P20))
```

```
## [1] 2.220446e-16
```

```
PD_raw <- P20[, "D"]
PD <- pmax(PD_raw, 0.0005) # set a floor for
```

```
data.frame(
  rating = names(PD),
  PD_raw = round(PD_raw, 6),
  PD = round(PD, 6),
  floored = PD_raw < 0.0005
)
```

	rating	PD_raw	PD	floored
AAA	AAA	0.000027	0.000500	TRUE
AA	AA	0.000114	0.000500	TRUE
A	A	0.002237	0.002237	FALSE
BBB	BBB	0.004118	0.004118	FALSE
BB	BB	0.013191	0.013191	FALSE
B	B	0.067963	0.067963	FALSE
CCC/C	CCC/C	0.279564	0.279564	FALSE
D	D	1.000000	1.000000	FALSE

2 Task

2.1

Determine the credit risk for the portfolio of 100 borrowers based on the Basel Standardized Approach. Proceed by using the existing ratings as if they were external ratings.

```
str(portfolio)
```

```
## Classes 'data.table' and 'data.frame':  100 obs. of  4 variables:
## $ Obligor      : num  1 2 3 4 5 6 7 8 9 10 ...
## $ Rating       : chr  "BBB" "BB" "BB" "BB" ...
## $ Industry     : num  1 3 2 2 1 1 1 3 3 3 ...
## $ Loan.Volume: num  169000 149000 850000 76000 174000 108000 122000 33000 22000 38000 ...
```

```
portfolio[, .N, by = Rating][order(-N)]
```

Rating	N
BBB	30
BB	27
A	17
B	9
AA	8
CCC/C	6
AAA	3

```
sum(portfolio$Loan.Volume)
```

```
## [1] 16576000
```

```
range(portfolio$Loan.Volume)
```

```
## [1] 21000 850000
```

```
portfolio[, .N, by = Industry]
```

Industry	N
1	42
3	29
2	29

```
rw_table <- data.table(
  Rating = c("AAA", "AA", "A", "BBB", "BB", "B", "CCC/C", "D"),
  RW = c(0.20, 0.20, 0.50, 1.00, 1.00, 1.50, 1.50, 1.50)
)

port_sa <- merge(portfolio, rw_table, by = "Rating", all.x = TRUE)

port_sa[, RWA := Loan.Volume * RW]
```

```
head(port_sa)
```

Rating	Obligor	Industry	Loan.Volume	RW	RWA
A	7	1	122000	0.5	61000
A	8	3	33000	0.5	16500
A	13	3	850000	0.5	425000
A	18	1	191000	0.5	95500
A	23	1	850000	0.5	425000
A	24	1	198000	0.5	99000

```
total_EAD <- sum(port_sa$Loan.Volume)
total_RWA <- sum(port_sa$RWA)
capital_req_SA <- 0.08 * total_RWA
```

```
port_sa[, .(
  N = .N,
  EAD = sum(Loan.Volume),
  RW = unique(RW),
  RWA = sum(RWA),
  Capital = 0.08 * sum(RWA)
), by = Rating][order(Rating)]
```

Rating	N	EAD	RW	RWA	Capital
A	17	3430000	0.5	1715000	137200
AA	8	1410000	0.2	282000	22560
AAA	3	279000	0.2	55800	4464
B	9	887000	1.5	1330500	106440
BB	27	4428000	1.0	4428000	354240
BBB	30	4821000	1.0	4821000	385680
CCC/C	6	1321000	1.5	1981500	158520

```
sum(!is.na(port_sa$RW))
```

```
## [1] 100
```

```
total_RWA / total_EAD
```

```
## [1] 0.881624
```

```
capital_req_SA / total_EAD
```

```
## [1] 0.07052992
```